

Introduction to C++ Programming

A Foundation Guide for Beginners

Welcome to your C++ journey! C++ is one of the most powerful and widely used programming languages. It serves as the backbone for competitive programming, high-performance game development, system software, and many of the world's leading tech infrastructures.

What is C++?

C++ is a compiled and statically typed programming language. Here is what that means for you as a developer:

Compiled Language: Your C++ code is converted into machine code before it runs. This makes C++ programs incredibly fast and efficient because no translation happens during execution.

Statically Typed: You must explicitly declare what type of data you are using (e.g., number, text). This helps catch errors at "compile-time" rather than while the user is running the program.

Structure of a C++ Program

Every C++ program follows a standardized structure. Let's look at a classic "Hello World" example.

```
#include <iostream>
using namespace std;

int main() {
```

```
cout << "Hello, World!";  
return 0;  
}
```

Line-by-Line Breakdown:

- **Preprocessor Directives:** `#include <iostream>` tells the compiler to include standard Input/Output tools required for functions like `cin` and `cout`.
- **Namespaces:** `using namespace std;` allows us to use standard library names directly (otherwise we'd have to write `std::cout`).
- **The main() Function:** Execution starts here. Every C++ program must have exactly one `main()` function. `return 0;` indicates the program finished successfully.

⚠ Important Rule: The Semicolon (;)

In C++, almost every statement must end with a semicolon. Missing one will result in a compilation error.

Input and Output (I/O)

Communication with the user is handled via input/output streams.

Output using `cout`

```
cout << "Welcome to C++!";
```

The `<<` operator "pushes" data to the display stream.

Input using `cin`

```
int age;  
cin >> age;
```

The `>>` operator "extracts" data from the keyboard input.

Variables and Data Types

Variables are containers for storing data values. In C++, you must define the type of data the variable holds.

Data Type	Size (typical)	Common Use Case
int	4 bytes	Whole numbers (e.g., 10, -500)
double	8 bytes	Decimal/Floating point values (e.g., 3.14)
char	1 byte	Single characters (e.g., 'A', '\$')
bool	1 byte	Boolean values (true or false)

Constants

If you have a value that should never change, use the `const` keyword. This makes your code safer and more readable.

```
const double PI = 3.14159;
```

Operators in C++

Arithmetic & Assignment

C++ uses standard symbols for math: `+`, `-`, `*`, `/`, and `%` (the modulo operator, used for finding remainders).

```
int x = 5 % 2; // Result: 1 (Remainder of 5 divided by 2)
x++;          // Shorthand for x = x + 1
```

Comparison Operators

Used to check conditions (`=` for equal, `≠` for not equal, `>`, `<`).

Control Flow (Decision Making)

Control flow allows your program to make choices based on data.

```
int age = 18;

if (age ≥ 18) {
    cout << "You are an adult";
} else {
    cout << "You are a minor";
}
```

Logical Operators

- `&&` (AND): True if both conditions are true.
- `||` (OR): True if at least one condition is true.
- `!` (NOT): Reverses the logical state.